

UI COMPONENT TECHNICAL SPECIFICATION (DESIGN SYSTEM)

1. Global System Conventions

1.1 Universal Props (Inherited by all components)

All components extend their corresponding native HTML attributes and support element polymorphism:

- `as`: Type `ElementType` (e.g., `'div'`, `'span'`, `'section'`). Underlying HTML element to render. Default value defined per component.
- `className`: Type `string`. Additional CSS classes for controlled overrides.
- `style`: Type `CSSProperties`. Inline styles for exceptional use cases.

1.2 Standardized Token Scales

Size Scale (`SizeScale`)

- Allowed values: `'xs'` | `'sm'` | `'md'` | `'lg'` | `'xl'`
- Standard default value: `'md'`

Spacing Scale (`SpacingScale`)

- Allowed values: `'none'` (0px) | `'xs'` (4px) | `'sm'` (8px) | `'md'` (16px) | `'lg'` (24px) | `'xl'` (32px) | `'2xl'` (48px)

Border Radius Scale (`RadiusScale`)

- Allowed values: `'none'` (0px) | `'xs'` (2px) | `'sm'` (4px) | `'md'` (8px) | `'lg'` (16px) | `'xl'` (24px) | `'full'` (9999px)

Semantic Color Palette (`SemanticColors`)

- Allowed values: `'primary'` | `'secondary'` | `'neutral'` | `'success'` | `'warning'` | `'danger'` | `'info'`

2. Typography & Base

2.1 Component: Text

Component for rendering plain text, paragraphs, or labels with typographical control.

Property	Type / Allowed Values	Required	Default Value	Description
<code>children</code>	<code>ReactNode</code>	Yes	-	Textual content to render.
<code>as</code>	<code>'p'</code> <code>'span'</code> <code>'label'</code> <code>'div'</code> <code>'small'</code>	No	<code>'p'</code>	Semantic HTML tag.

variant	'regular' 'mono' 'display'	No	'regular'	Font family assigned in the theme.
size	SizeScale ('xs' 'sm' 'md' 'lg' 'xl')	No	'md'	Font size (12px, 14px, 16px, 18px, 20px).
weight	400 500 600 700	No	400	Font weight (Regular, Medium, Semi-Bold, Bold).
italic	boolean (true false)	No	false	Applies italic font style.
underline	boolean (true false)	No	false	Applies text underline.
strikethrough	boolean (true false)	No	false	Applies strikethrough line.
color	SemanticColors 'dimmed' 'inherit'	No	'inherit'	Semantic color token or dimmed text.
align	'left' 'center' 'right' 'justify'	No	'left'	Text alignment.
truncate	boolean number	No	false	true for single-line truncation with ..., or an integer from 1 to 5 for line clamping.

2.2 Component: Title

Semantic component for structured headings.

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	Yes	-	Heading content.
order	1 2 3 4 5 6	No	1	HTML heading level (<h1> to <h6>).
size	'h1' 'h2' 'h3' 'h4' 'h5' 'h6'	No	Matches order	Allows changing visual size without breaking HTML tag hierarchy.
weight	500 600 700 800	No	700	Heading font weight.
color	SemanticColors 'inherit'	No	'inherit'	Color token.

2.3 Component: Divider

Visual and structural separator line.

Property	Type / Allowed Values	Required	Default Value	Description
orientation	'horizontal' 'vertical'	No	'horizontal'	Divider rendering orientation.
variant	'solid' 'dashed' 'dotted'	No	'solid'	Stroke style.

size	1 2 4 8	No	1	Line thickness in pixels.
color	SemanticColors 'border'	No	'border'	Color token applied to the line.
label	string ReactNode	No	undefined	Optional centered text on the divider line.

3. Structure & Layout Orchestrators

3.1 Component: Box (Surface)

Polymorphic base container with design token access.

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	No	-	Contained child elements.
size	'xs' 'sm' 'md' 'lg' 'xl' 'full'	No	undefined	Sets a predefined max-width limit. If omitted, takes full parent flow width.
centered	boolean (true false)	No	false	Applies automatic horizontal centering (margin-inline: auto).

bg	'app' 'surface' 'elevated' SemanticColors	No	'surface'	Contextual background color based on active theme.
padding	SpacingScale	No	'none'	Uniform inner padding based on spacing tokens.
radius	RadiusScale	No	'none'	Corner rounding based on radius tokens.
border	boolean	No	false	Applies subtle perimeter border using the border token.
shadow	SizeScale 'none'	No	'none'	Elevation level via multi-layer shadow tokens.

3.2 Component: Stack (VStack)

Vertical layout orchestrator with uniform Flexbox spacing.

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	Yes	-	Child elements to stack vertically.
gap	SpacingScale	No	'md'	Vertical gap between child

				elements.
align	'stretch' 'flex-start' 'center' 'flex-end'	No	'stretch'	Cross-axis alignment (align-items).
justify	'flex-start' 'center' 'flex-end' 'space-between'	No	'flex-start'	Main-axis distribution (justify-content).

3.3 Component: Group (HStack)

Horizontal layout orchestrator with line-wrapping controls.

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	Yes	-	Child elements to align horizontally.
gap	SpacingScale	No	'md'	Horizontal gap between child elements.
align	'stretch' 'flex-start' 'center' 'flex-end'	No	'center'	Cross-axis vertical alignment.
justify	'flex-start' 'center' 'flex-end' 'space-between' 'space-around'	No	'flex-start'	Main-axis horizontal distribution.

wrap	'nowrap' 'wrap' 'wrap-reverse'	No	'wrap'	Wrapping behavior on container overflow.
grow	boolean (true false)	No	false	If true, child elements grow equally to fill available width.

3.4 Component: Grid and Grid.Col

12-column CSS Grid system.

Grid (Container)

Property	Type / Allowed Values	Required	Default Value	Description
columns	Integer between 1 and 24	No	12	Total number of grid columns.
gutter	SpacingScale	No	'md'	Spacing between grid rows and columns.

Grid.Col (Grid Item)

Property	Type / Allowed Values	Required	Default Value	Description
span	Integer between 1 and 12 'auto'	No	12	Column span at base viewport resolution.

sm / md / lg / xl	Integer between 1 and 12 'auto'	No	undefined	Column span across responsive breakpoints.
offset	Integer between 1 and 11	No	0	Left column offset margin.

3.5 Component: AspectRatio

Container locking a mathematical aspect ratio.

Property	Type / Allowed Values	Required	Default Value	Description
ratio	Positive float (e.g., 16/9 $\approx$ 1.777, 4/3, 1)	Yes	1	Aspect ratio (width / height).
children	ReactNode	Yes	-	Inner content constrained by the aspect ratio.

3.6 Component: Container

Centered horizontal width restrictor.

Property	Type / Allowed Values	Required	Default Value	Description
size	'xs' (540px) 'sm' (720px) 'md' (960px) 'lg' (1140px) 'xl' (1320px) 'fluid' (100%)	No	'md'	Maximum container width (max-width).

padding	SpacingScale	No	'md'	Safe horizontal padding to prevent viewport edge overflow.
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4. Actions & Navigation

4.1 Component: Button

Primary interactive action element.

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	Yes	-	Button label or inner elements.
variant	'filled' 'light' 'outline' 'subtle' 'link'	No	'filled'	Visual style and hierarchy.
color	SemanticColors	No	'primary'	Semantic color palette.
size	SizeScale	No	'md'	Height, padding, and font scale.
radius	RadiusScale	No	'md'	Corner border radius.
loading	boolean (true false)	No	false	Displays spinner and disables interaction.
disabled	boolean (true false)	No	false	Disables interactive

				state and applies opacity.
fullWidth	boolean (true false)	No	false	Sets width to 100% of parent container.
leftSection	ReactNode	No	undefined	Icon or element rendered left of text.
rightSection	ReactNode	No	undefined	Icon or element rendered right of text.
onClick	(event: MouseEvent) => void	No	undefined	Click event callback.

4.2 Component: IconButton

Square or circular button optimized exclusively for rendering an icon.

Property	Type / Allowed Values	Required	Default Value	Description
icon	ReactNode	Yes	-	SVG/Icon element to render.
aria-label	string	Yes	-	Required screen reader accessibility label.

variant	'filled' 'light' 'outline' 'subtle'	No	'subtle'	Visual style.
color	SemanticColors	No	'neutral'	Semantic color palette.
size	SizeScale	No	'md'	Fixed dimension (e.g., 'md' = 36x36px).
radius	RadiusScale	No	'md'	Corner radius ('full' for circular).
loading	boolean	No	false	Loading state.
disabled	boolean	No	false	Disabled state.

5. Data Capture (Forms)

5.1 Shared Base Wrapper: InputWrapper

Integrated across all form input components:

Common Property	Type / Allowed Values	Required	Default Value	Description
label	string ReactNode	No	undefined	Top input label text.
description	string ReactNode	No	undefined	Helper text below label.
error	string boolean	No	undefined	Displays error message string and triggers danger state.

required	boolean (true false)	No	false	Adds red asterisk and native HTML required attribute.
size	SizeScale	No	'md'	Input dimension scale.
disabled	boolean	No	false	Disables user data entry.

5.2 Component: TextField

Single-line text input field.

Property	Type / Allowed Values	Required	Default Value	Description
value	string	No	undefined	Controlled input value.
defaultValue	string	No	undefined	Initial uncontrolled value.
placeholder	string	No	undefined	Temporary placeholder text.
type	'text' 'password' 'email' 'search' 'tel' 'url'	No	'text'	Native HTML input type.
leftSection	ReactNode	No	undefined	Decorative icon inside input on the

				left.
rightSection	ReactNode	No	undefined	Action icon/button inside input on the right (e.g., password toggle).
onChange	(event: ChangeEvent<HTMLInputElement>) => void	No	undefined	Change event callback.

5.3 Component: NumberInput

Controlled numeric input with increment/decrement step controls.

Property	Type / Allowed Values	Required	Default Value	Description
value	number "	No	undefined	Controlled numeric value.
min	Real number (e.g., -100, 0)	No	-Infinity	Minimum allowed value.
max	Real number (e.g., 100, 9999)	No	Infinity	Maximum allowed value.
step	Positive real number (e.g., 1, 0.5, 0.01)	No	1	Increment/decrement step.
precision	Integer between 0 and 6	No	0	Mandatory decimal places formatted in

				display.
hideControls	boolean (true false)	No	false	Hides built-in up/down spinner arrows.
onChange	(value: number undefined) => void	No	undefined	Emits parsed pure numeric value.

5.4 Component: Select

Single-option dropdown selector.

Property	Type / Allowed Values	Required	Default Value	Description
data	Array<{ label: string, value: string, disabled?: boolean }> string[]	Yes	-	List of options to render.
value	string	No	undefined	Controlled selected value.
placeholder	string	No	undefined	Placeholder text when unselected.
searchable	boolean (true false)	No	false	Enables text filtering in dropdown menu.
clearable	boolean (true false)	No	false	Displays 'x' button to clear current

				selection.
onChange	(value: string null) => void	No	undefined	Option selection callback.

5.5 Component: Switch

Binary toggle switch.

Property	Type / Allowed Values	Required	Default Value	Description
checked	boolean (true false)	No	undefined	Controlled binary state.
label	string ReactNode	No	undefined	Label text alongside toggle.
labelPosition	'left' 'right'	No	'right'	Side to render label relative to toggle track.
color	SemanticColors	No	'primary'	Background track color when active (checked=true).
size	SizeScale	No	'md'	Track capsule and inner knob scale.
onChange	(event: ChangeEvent<HTMLInputElement>) => void	No	undefined	State change callback.

6. Data Display & Media

6.1 Component: Avatar

Graphical element for user identity display.

Property	Type / Allowed Values	Required	Default Value	Description
src	string (URL)	No	undefined	Remote image resource URL.
name	string (e.g., "Sebastián Narváez")	No	undefined	Generates fallback initials (e.g., "SN") if src is missing or fails.
alt	string	No	'avatar'	Accessibility alt text.
size	SizeScale Pixel integer (e.g., 48)	No	'md'	Avatar dimension.
radius	RadiusScale	No	'full'	Corner radius ('full' for circular).
color	SemanticColors 'auto'	No	'neutral'	Fallback background color when initials are displayed.

6.2 Component: Image

Optimized `<img>` element with loading and fallback handling.

Property	Type / Allowed Values	Required	Default Value	Description
src	string (URL)	Yes	-	Main image resource path.
alt	string	Yes	-	Mandatory accessibility alt text.
fit	'cover' 'contain' 'fill' 'scale-down' 'none'	No	'cover'	CSS object-fit rule.
fallback	ReactNode	No	undefined	Custom component or element to render on image error (e.g., <Box>, <Text>, or icon).
radius	RadiusScale	No	'none'	Corner border radius.
loading	'lazy' 'eager'	No	'lazy'	Native browser loading strategy.

6.3 Component: Badge

Pill indicator for statuses, counts, and categories.

Property	Type / Allowed Values	Required	Default Value	Description
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children	ReactNode	Yes	-	Badge label content.
variant	'filled' 'light' 'outline' 'dot'	No	'light'	Visual style ('dot' adds leading indicator circle).
color	SemanticColors	No	'primary'	Semantic color token.
size	SizeScale	No	'md'	Height and padding scale.
radius	RadiusScale	No	'xl'	Container border radius.
leftSection	ReactNode	No	undefined	Leading element or icon.

6.4 Component: Card

Modular compound container (Card.Header, Card.Body, Card.Footer).

Property	Type / Allowed Values	Required	Default Value	Description
children	ReactNode	Yes	-	Inner card contents.
padding	SpacingScale	No	'md'	Uniform internal padding.
radius	RadiusScale	No	'md'	Corner border radius.

withBorder	boolean (true false)	No	true	Enables subtle delimiting perimeter border.
shadow	SizeScale 'none'	No	'sm'	Elevation shadow level.

7. Overlays & Feedback

7.1 Component: Modal

Portal-based dialog overlay blocking underlying DOM interaction.

Property	Type / Allowed Values	Required	Default Value	Description
opened	boolean (true false)	Yes	-	React visibility state.
onClose	() => void	Yes	-	Callback executed upon requesting modal close.
title	string ReactNode	No	undefined	Title rendered in modal header.
size	'xs' (320px) 'sm' (440px) 'md' (560px) 'lg' (720px) 'xl' (960px) 'full' (100vw/vh)	No	'md'	Maximum modal window width.
centered	boolean (true false)	No	true	Centers modal vertically and horizontally on

				screen.
closeOnClickOutside	boolean (true false)	No	true	Closes modal when clicking backdrop area.
closeOnEscape	boolean (true false)	No	true	Closes modal on Escape keypress.
withCloseButton	boolean (true false)	No	true	Renders top-right close "X" button.

7.2 Component: Tooltip

Contextual floating text on hover or focus.

Property	Type / Allowed Values	Required	Default Value	Description
label	string ReactNode	Yes	-	Explanatory tooltip text.
children	ReactElement	Yes	-	Target anchor element for tooltip positioning.
position	'top' 'right' 'bottom' 'left'	No	'top'	Placement side relative to anchor element.
openDelay	Positive integer in ms (0 to 2000)	No	0	Delay before showing tooltip.

closeDelay	Positive integer in ms (0 to 2000)	No	0	Delay before hiding tooltip.
withArrow	boolean (true false)	No	true	Renders triangular pointer arrow towards anchor.

7.3 Component: Loader

Processing and activity status indicator.

Property	Type / Allowed Values	Required	Default Value	Description
variant	'spinner' 'dots' 'bars'	No	'spinner'	SVG animation style.
color	SemanticColors	No	'primary'	Stroke color token.
size	SizeScale Pixel integer (12 to 120)	No	'md'	Indicator dimensions.

8. Technical Implementation Guidelines (Developer Instructions)

1. **Strict Typings:** Every prop interface must be exported with the [ComponentName]Props naming convention (e.g., export interface ButtonProps extends BaseComponentProps, ...).
2. **CSS Modules:** No .tsx component file should use inline styles except for dynamic CSS variable calculations. All styling must reside within its respective [ComponentName].module.css file.
3. **ForwardRef:** All interactive and form controls (Button, TextField, Select, IconButton) must implement React.forwardRef to pass through references to

underlying DOM nodes.

8.1 Image Component Internal Implementation:

1. Maintain local `hasError` state (initialized to `false`).
2. Listen to native `onError` event on the `<img>` tag; update `hasError` to `true` when triggered.
3. If `hasError` is `true` and the `fallback` prop is supplied, render the custom fallback node inside the container matching the specified image dimensions.